Klinikum rechts der Isar
Technische Universität München

# MICCAI 2024

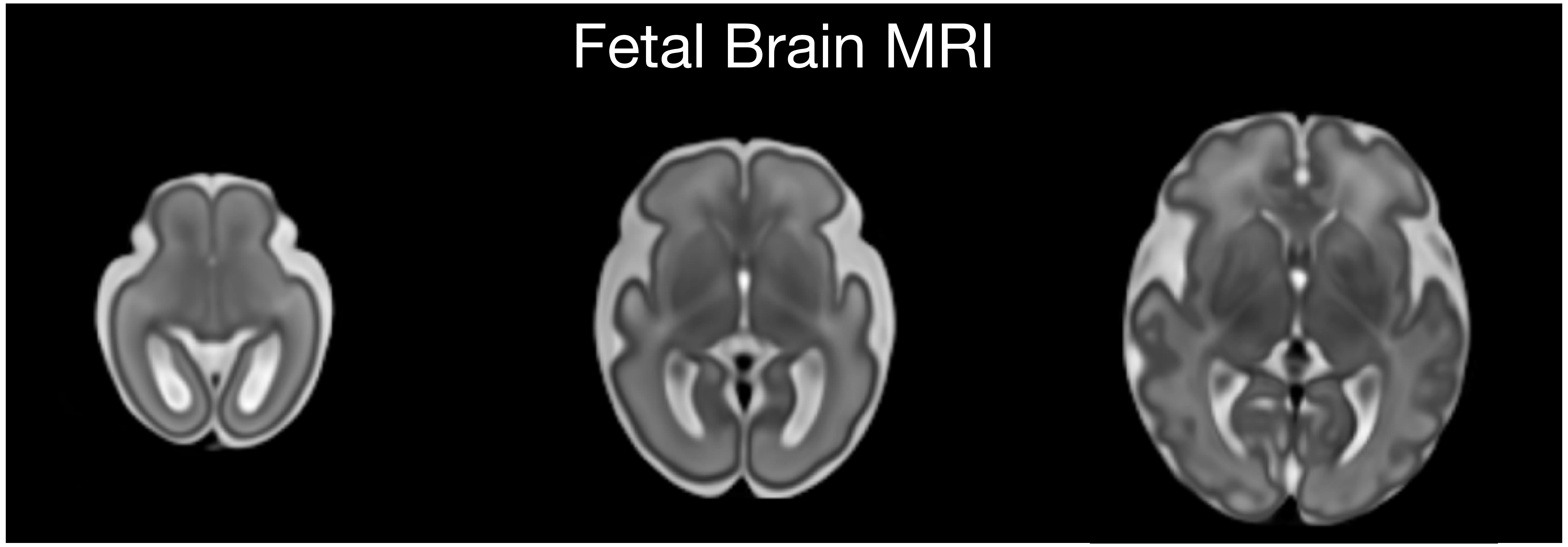
# INR Tutorial

Maik Dannecker, 06.10.2024

MICCAI 2024 Marrakesh MOROCCO

# Fetal Brain Development

**Fetal brains** undergo **rapid changes** throughout pregnancy involving different stages of growth and cortical folding. Understanding the spatio-temporal trajectory of fetal brain development is crucial for early identification of atypical brain patterns and potential neurodevelopmental disorders.




23 weeks 30 weeks 37 weeks

—> There is a need for spatio-temporal representations of the fetal brain.

# Atlases in Medical Imaging

- In medical imaging, an atlas serves as a ***normative* representation** of an anatomy (e.g. brain)
- Use-cases?
    - Reference for spatial alignment (registration)
    - Prior knowledge (e.g., for segmentation of brain tissue)
    - Group analysis (e.g., **identification of atypical brain patterns**)

# (Classic) Atlas Construction

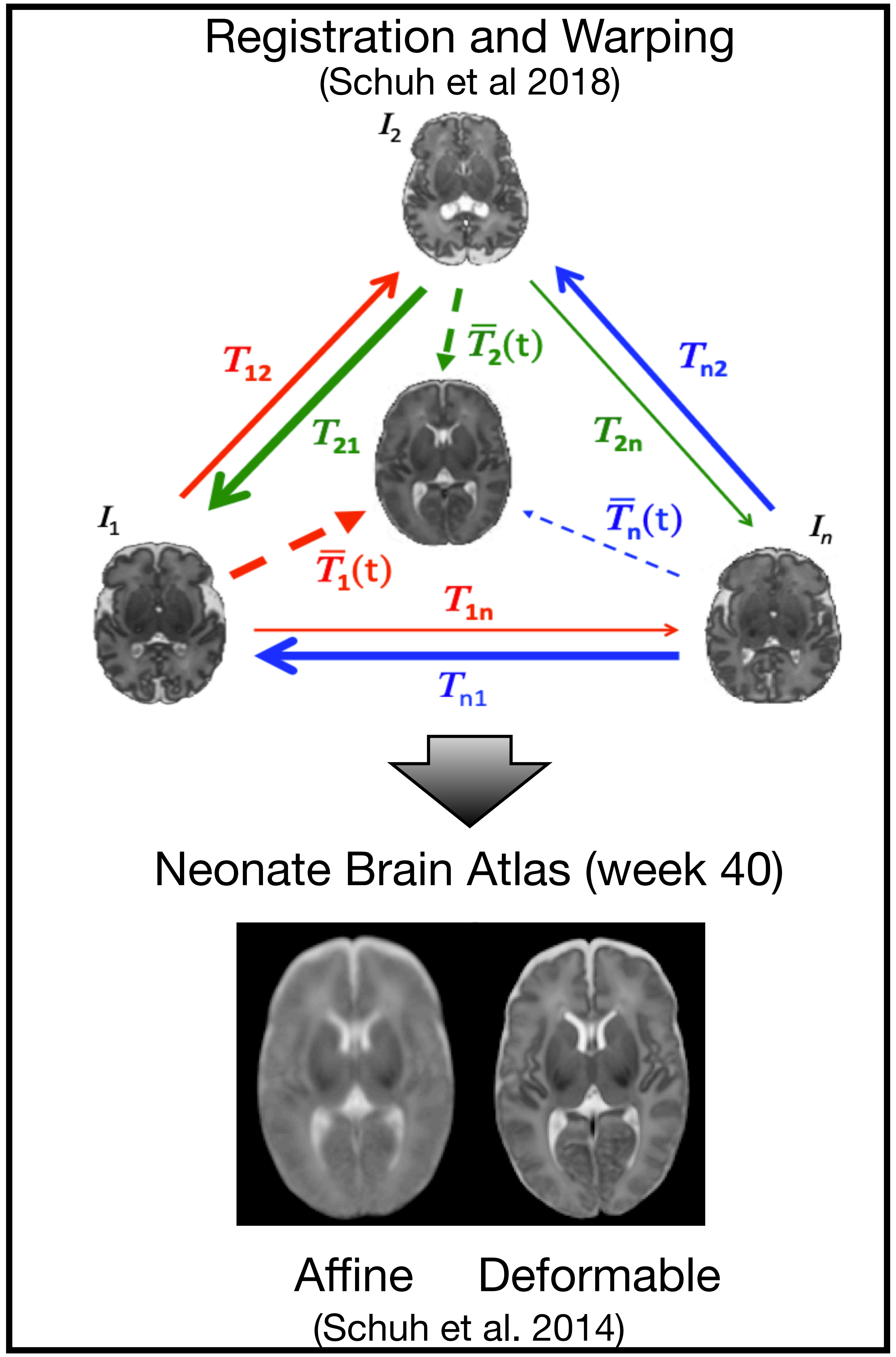
Atlas construction from a (normative) population:

- Trivial approach:
  - Mean over (aligned) images

    —> blurry due to high variability of images

- Registration approach:
  - Align images via affine or deformable registration
  - Apply *mean transformation* to warp images to common shape

    —> sharper delineation (variability in transformation not in the image)

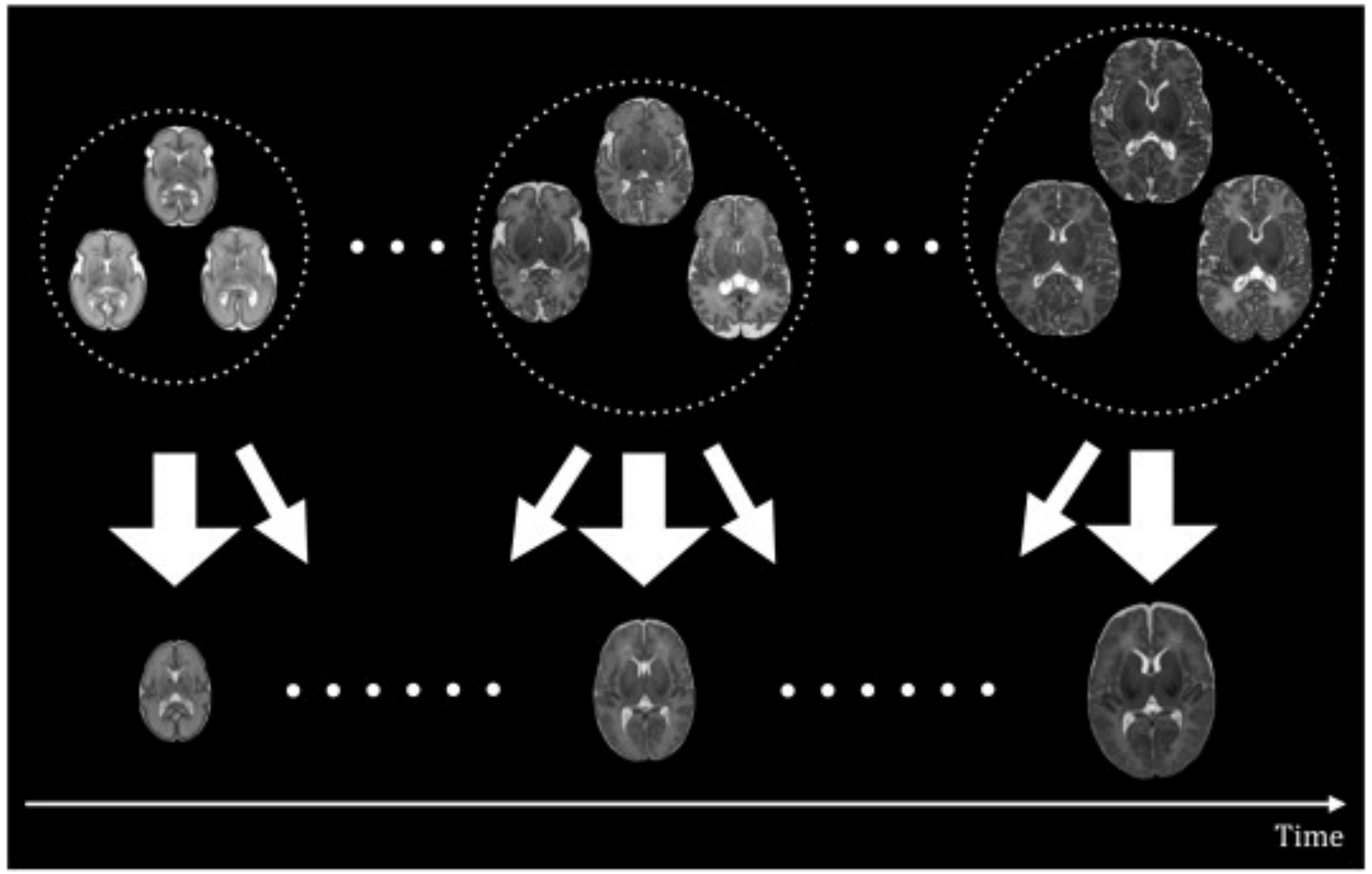


# Fetal Brain Atlas

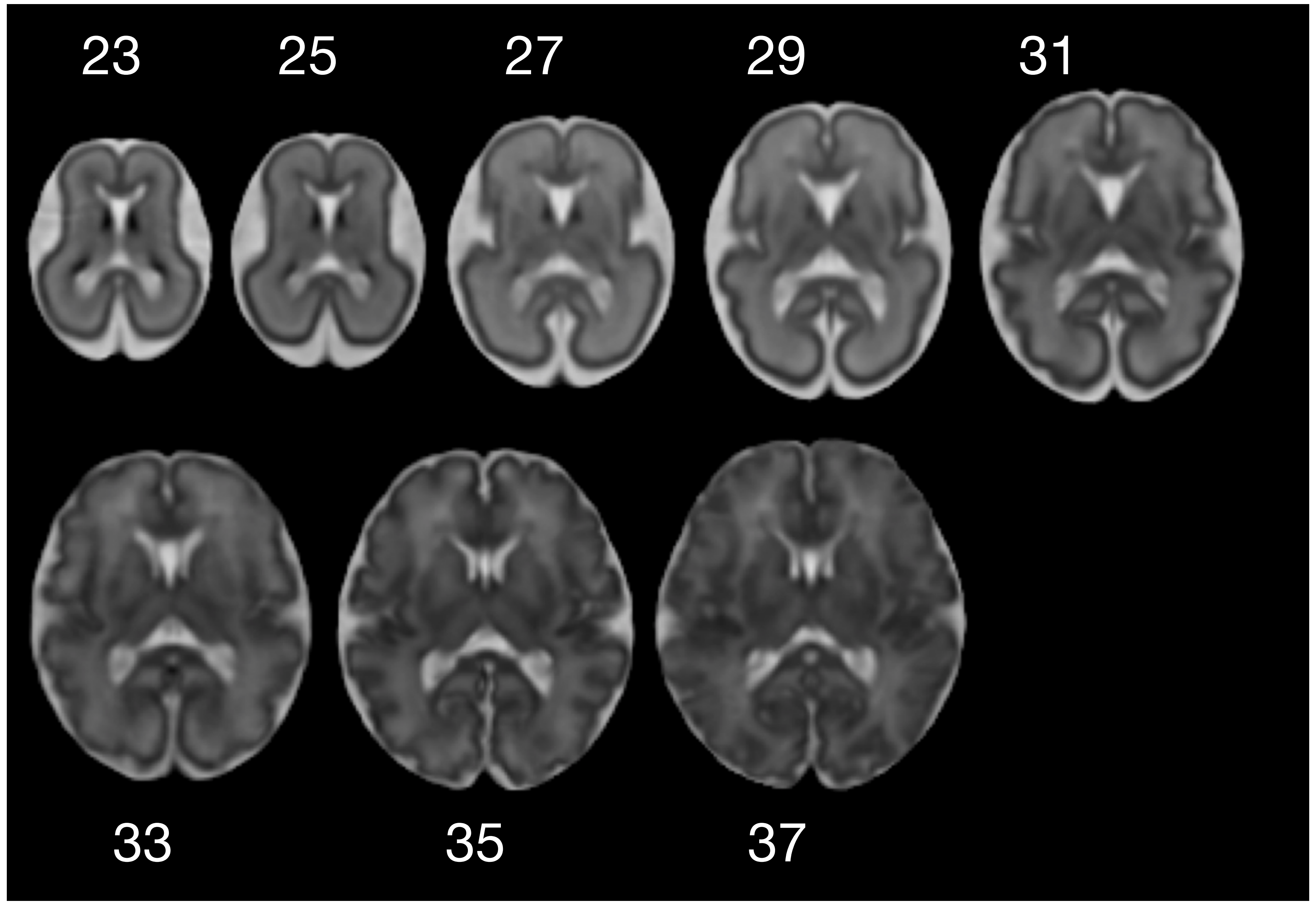
- Fetal brain MRI ranges from ~ 20 weeks gestational age (GA) to 38 weeks GA

  —> rapid brain development requires a dynamic atlas

  —> dynamic atlas requires sufficient data over time




(Serag et al. 2012)

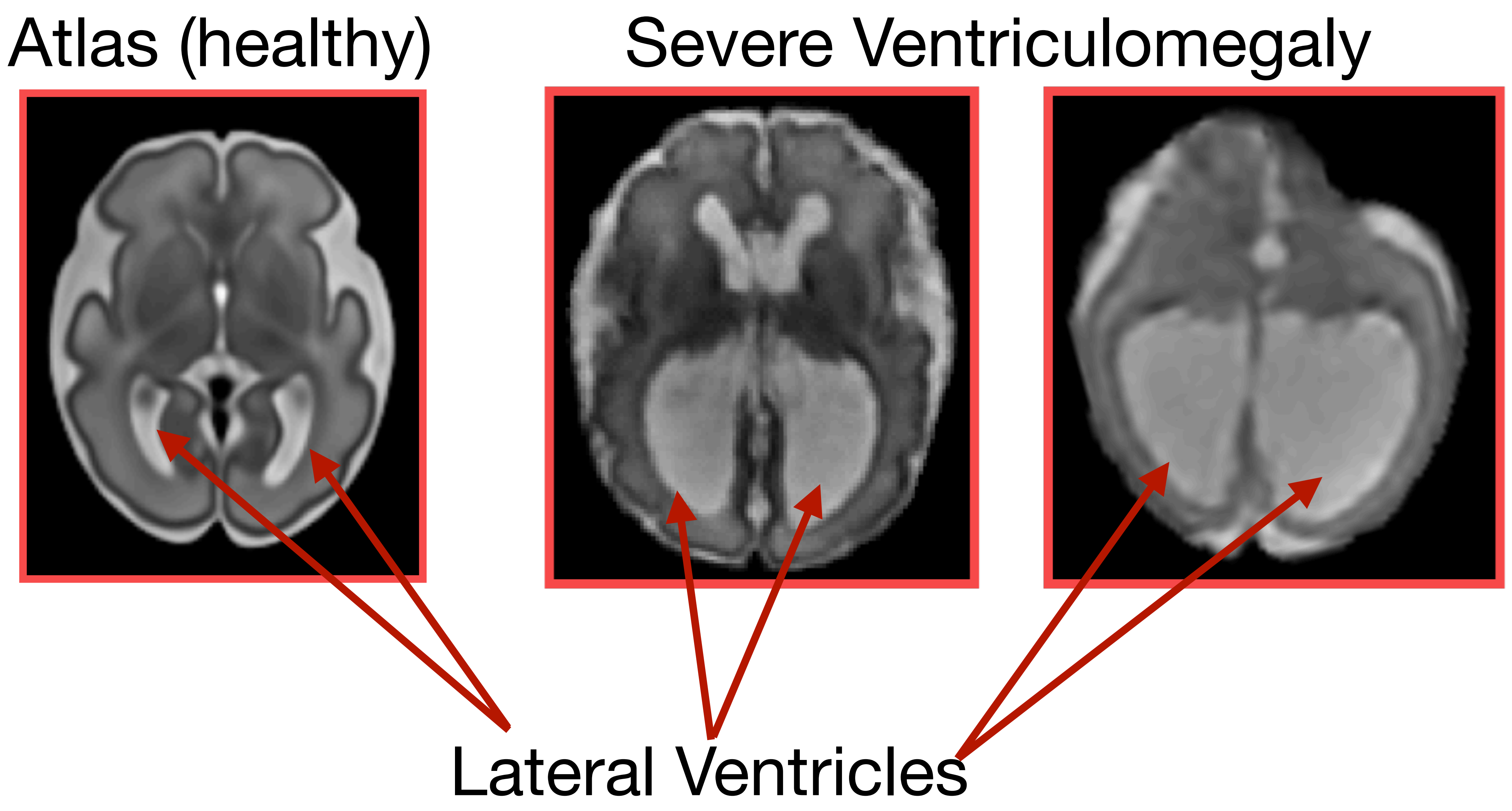



(brain-development.org (modified))

- Here, 80 subjects across age range used for atlas generation

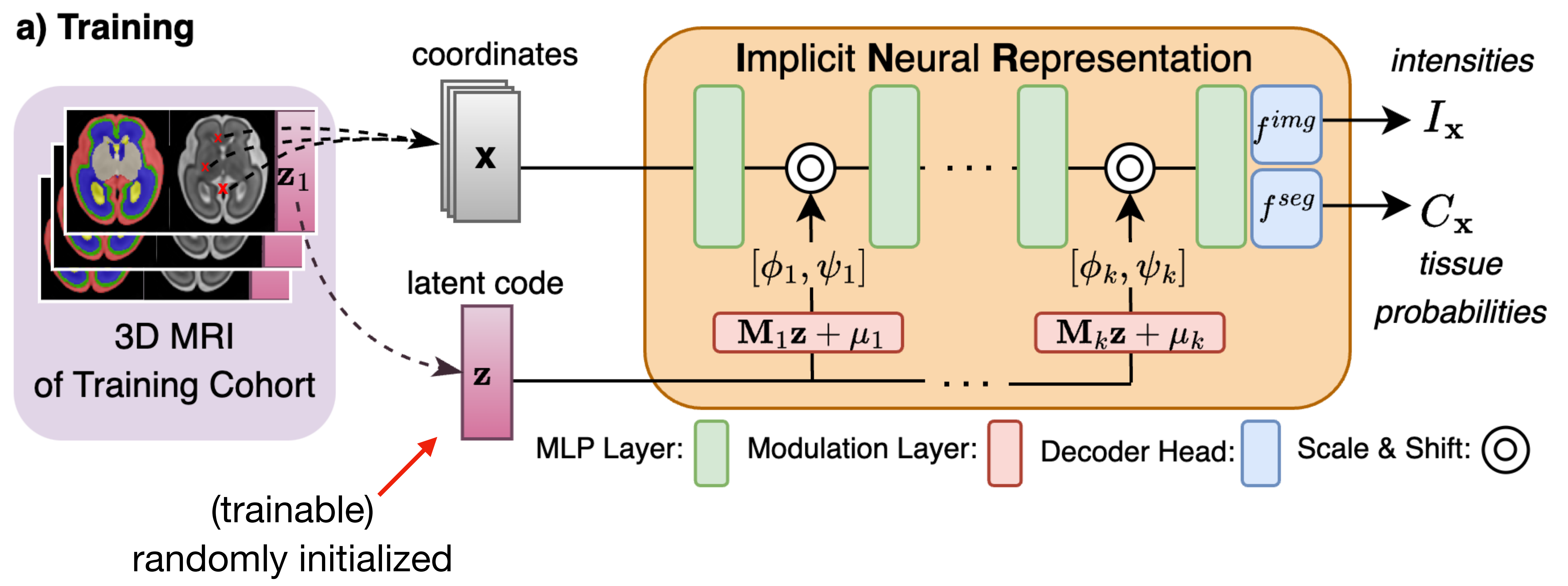
# Fetal Brain Atlas for Pathological Data

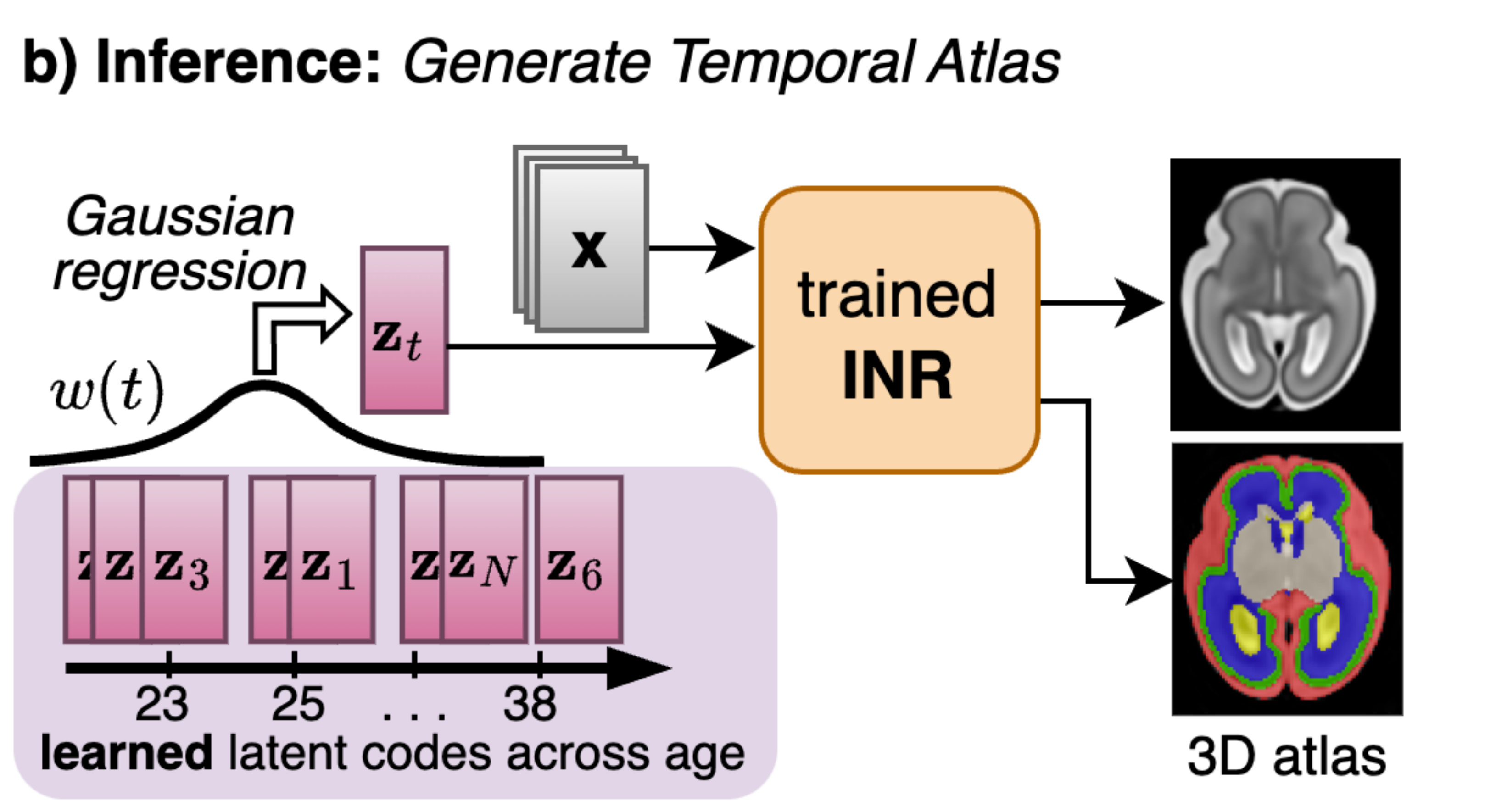
- Most atlases focus on healthy data and do not adequately model pathologies such as Ventriculomegaly (VM)
- Classic methods are static models.
  —> A new atlas is required for each use case.
- Public pathological data is rare, impeding atlas construction for pathology modeling. E.g., FeTA Dataset ~30 usable pathological scans, ~15 with VM.



# CINA: Conditional Implicit Neural Atlas

- Spatio-temporal atlas construction using conditional implicit neural representation.

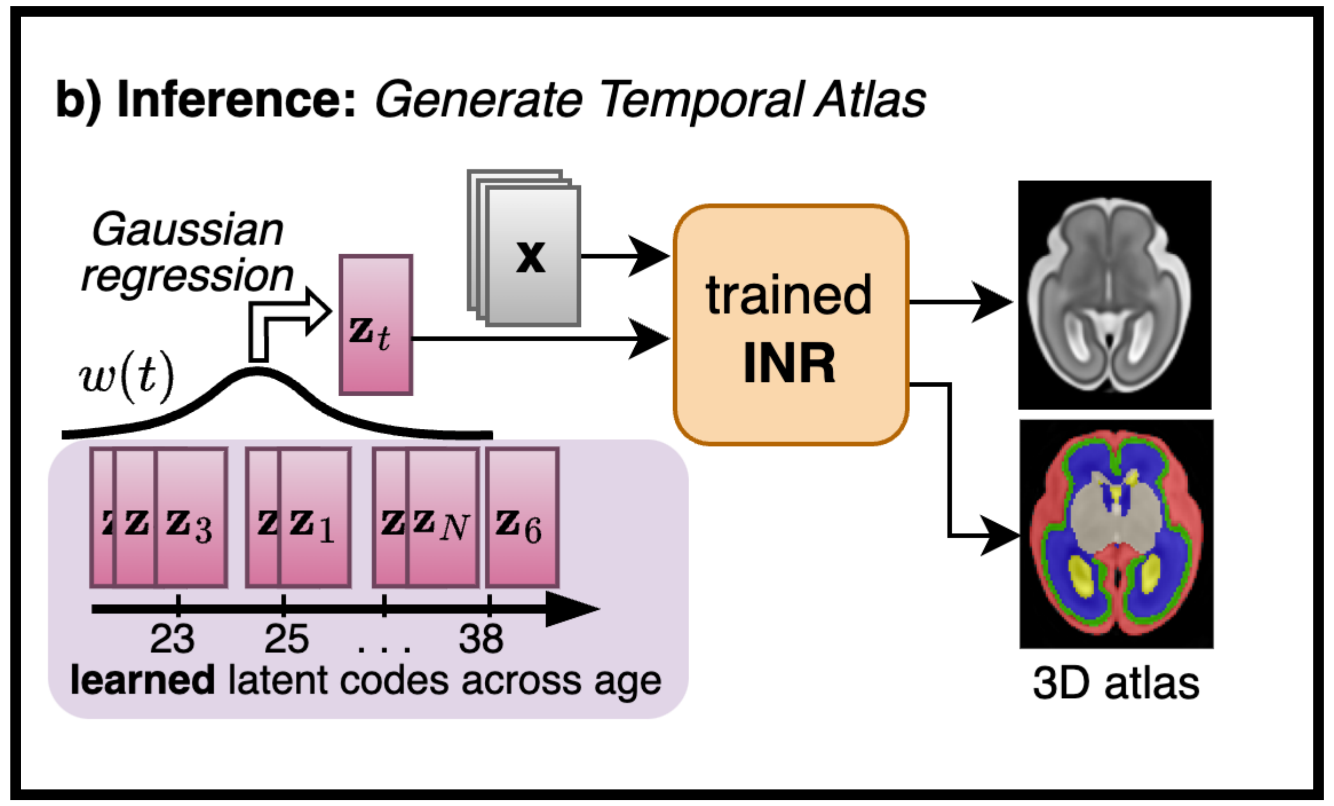


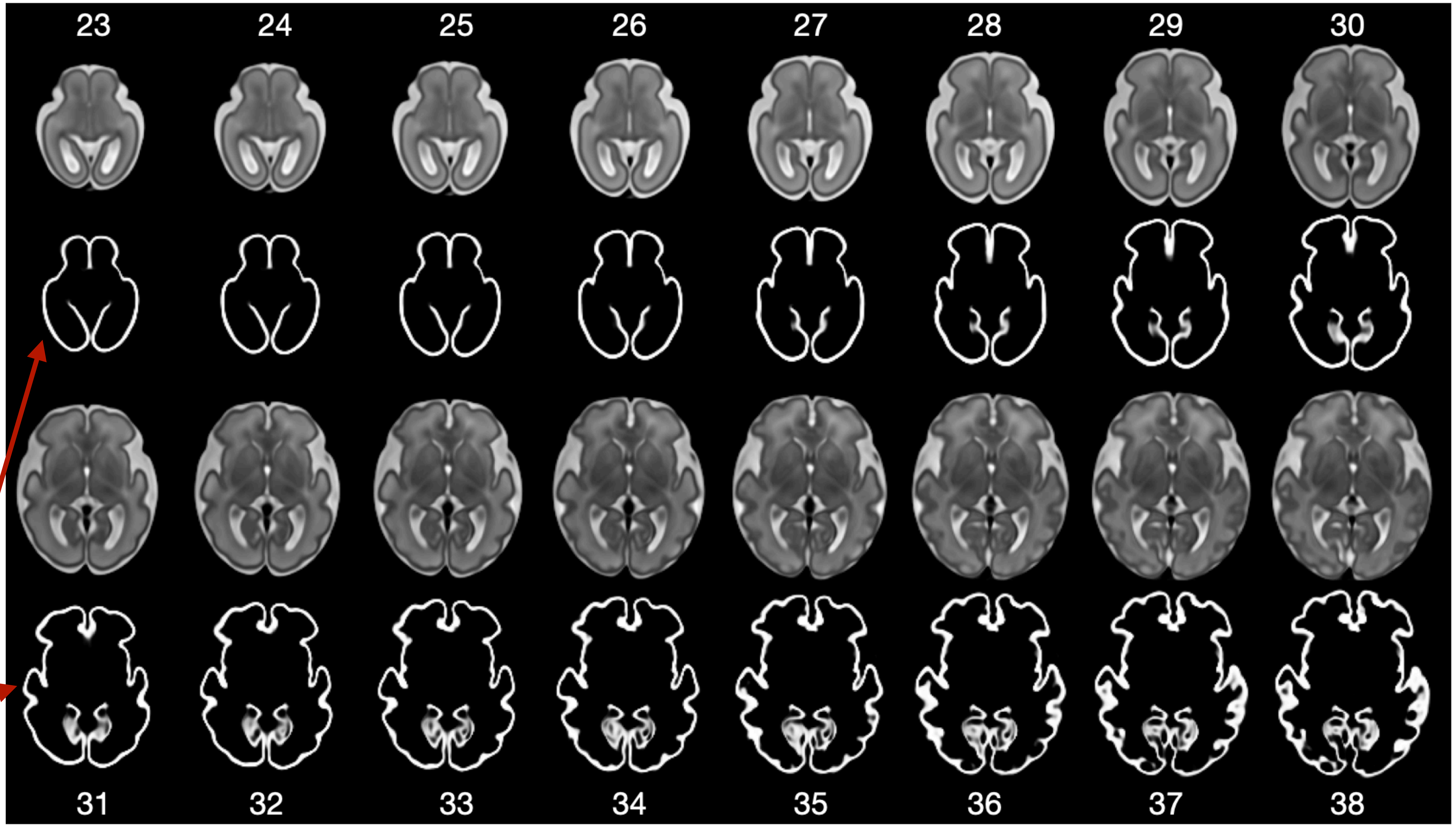




- Subject specific information (variability) is encoded in **z.**
- General brain representation encoded in the **INR.**

Dannecker et al. 2024 (preprint)

# Spatio-Temporal Atlas Construction with CINA



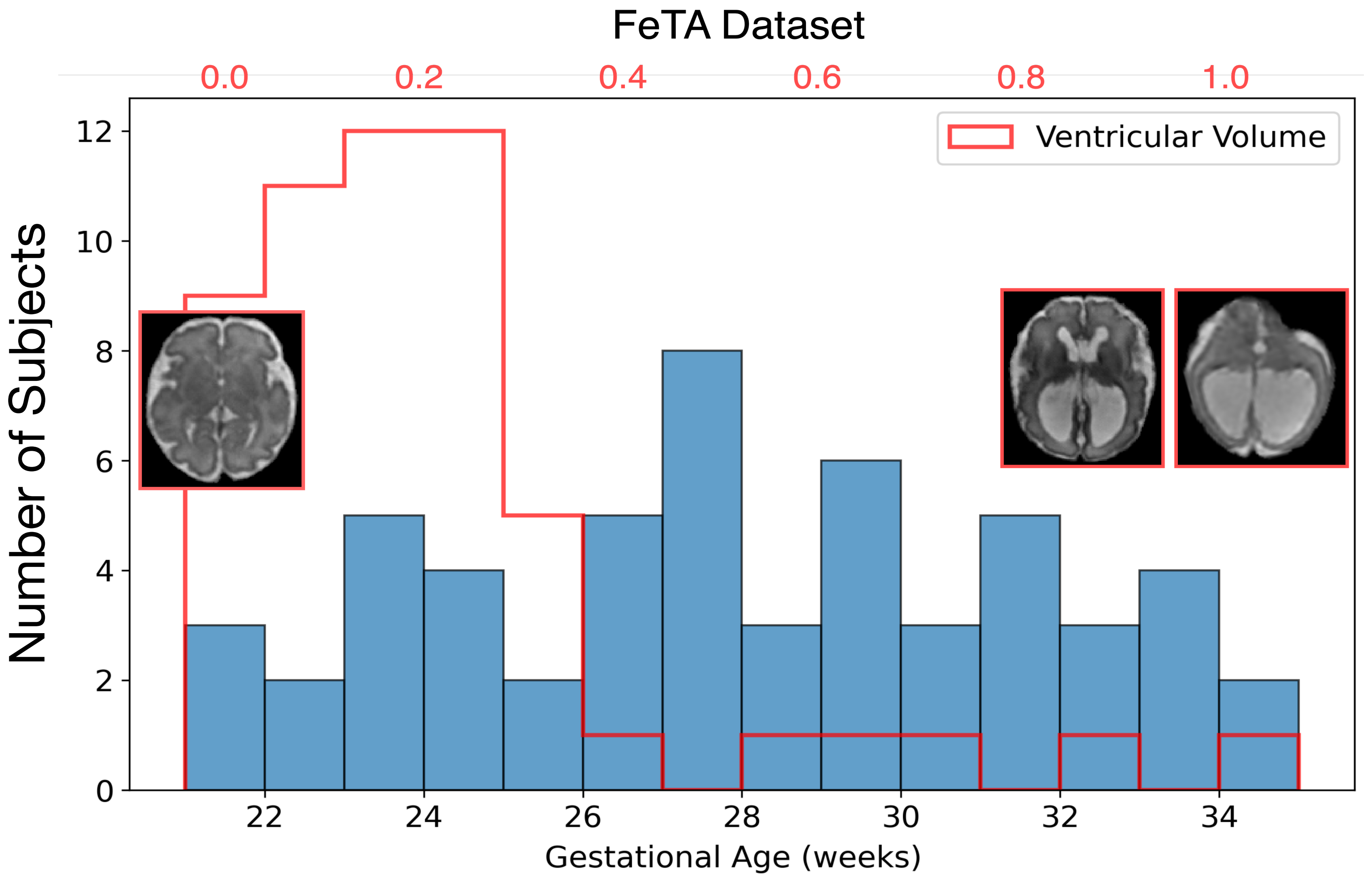

tissue probability map
(cortical grey matter)

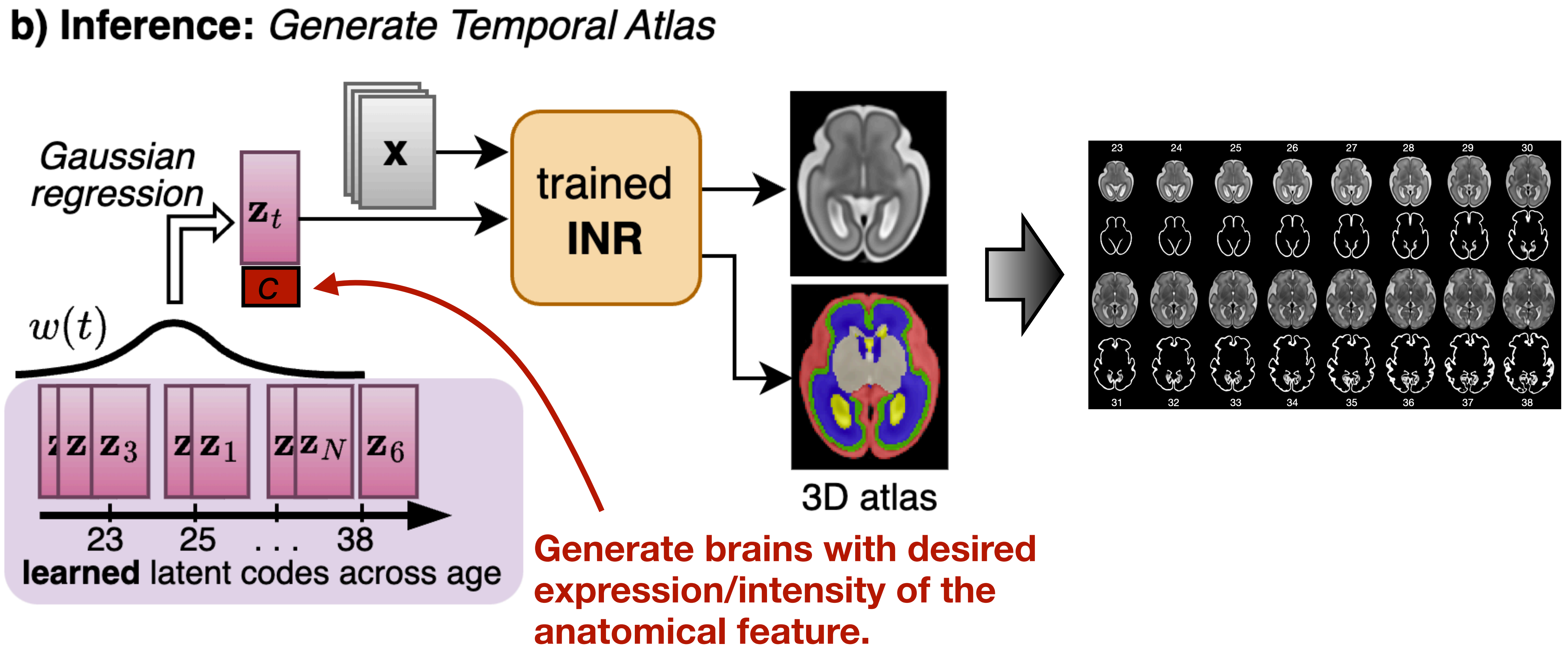
Fetal brain atlas of continuous spatial and temporal resolution.

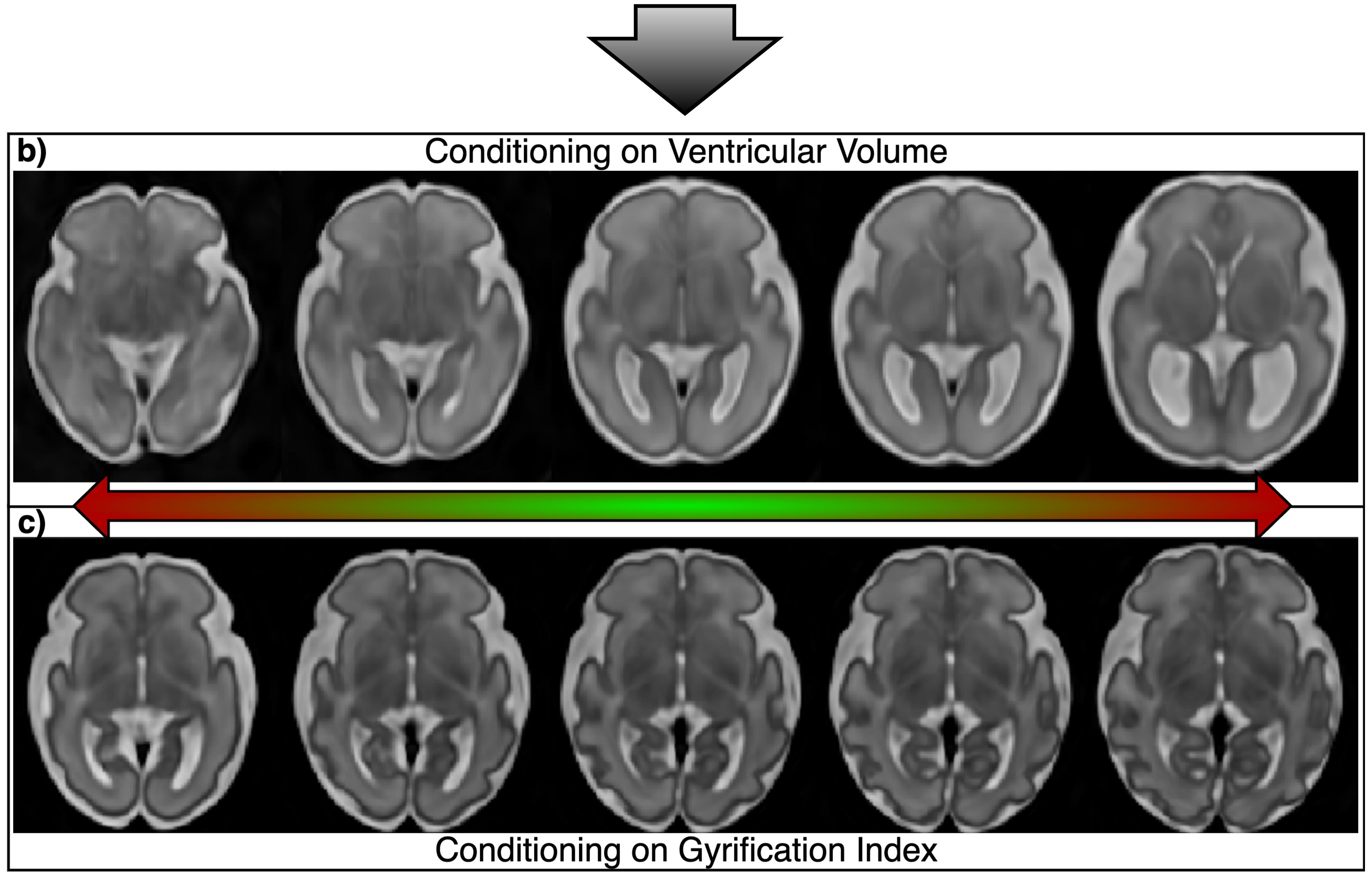
# Conditioning on Anatomical Features

Explicitly condition on:

- Ventircular Volume
  - Add subject's ventricular volume as additional dimension to **z** during training.
- Gyrification
  - Add subject's gyrification index (degree of cortical folding) to **z.**



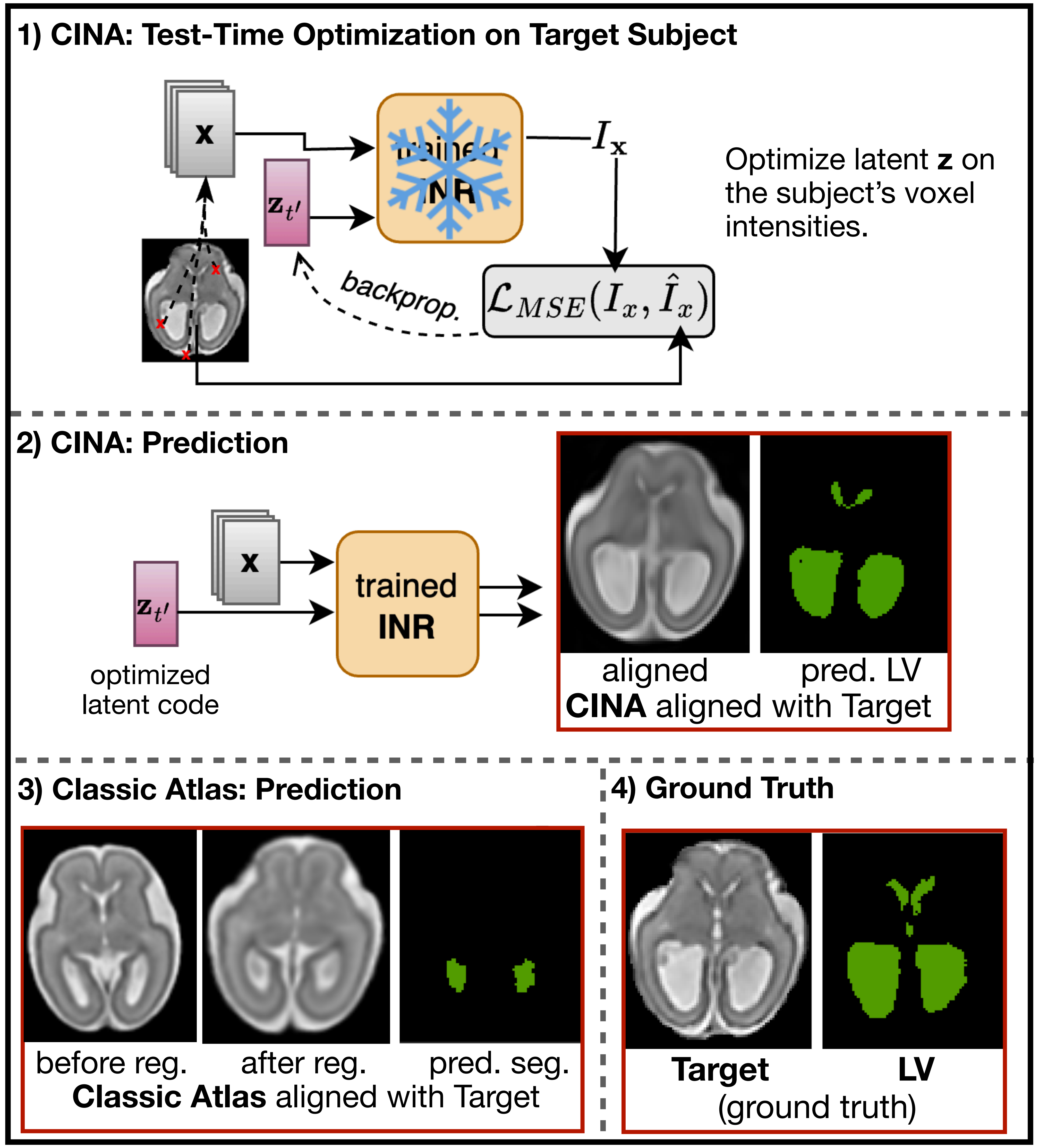
# Align Atlas to New Subjects

**CINA:**

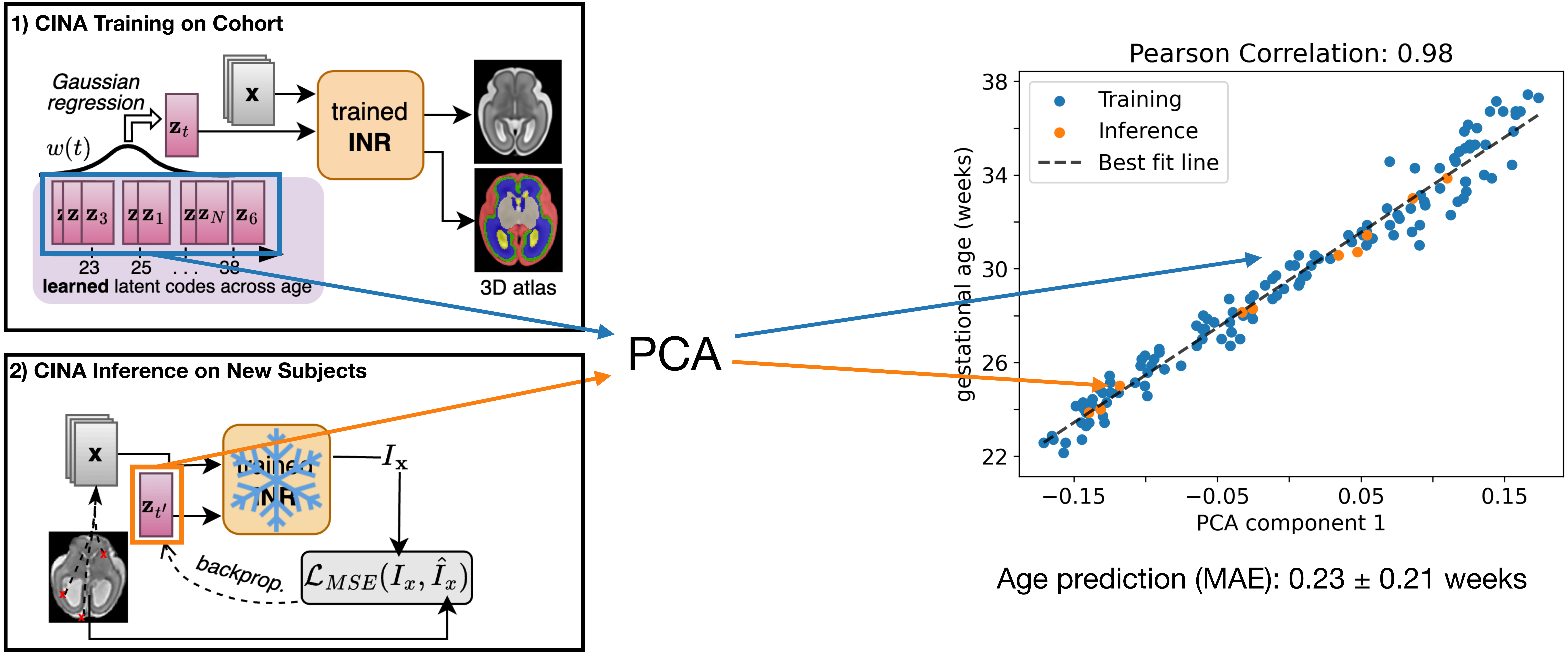
- initialize new latent code ***z***.
- optimize ***z*** during test-time, freezing the INR and minimizing MSE-Loss on voxel intensities of new subject.
- predict brain tissue via single forward pass

**Classic methods:**

- register atlas to new subject by finding optimal transformation *T*.
- project atlas tissue maps to new subject using *T*.



# Latent Representation




Age prediction (MAE): 0.23 ± 0.21 weeks

# Conclusion

**Classic Atlases:**

- capture **discrete time-points** (weeks) with **discrete spatial-resolution**.
- constitute a **static representation** of the (healthy) fetal brain.
- often **fail to** adequately **model pathological brains**.

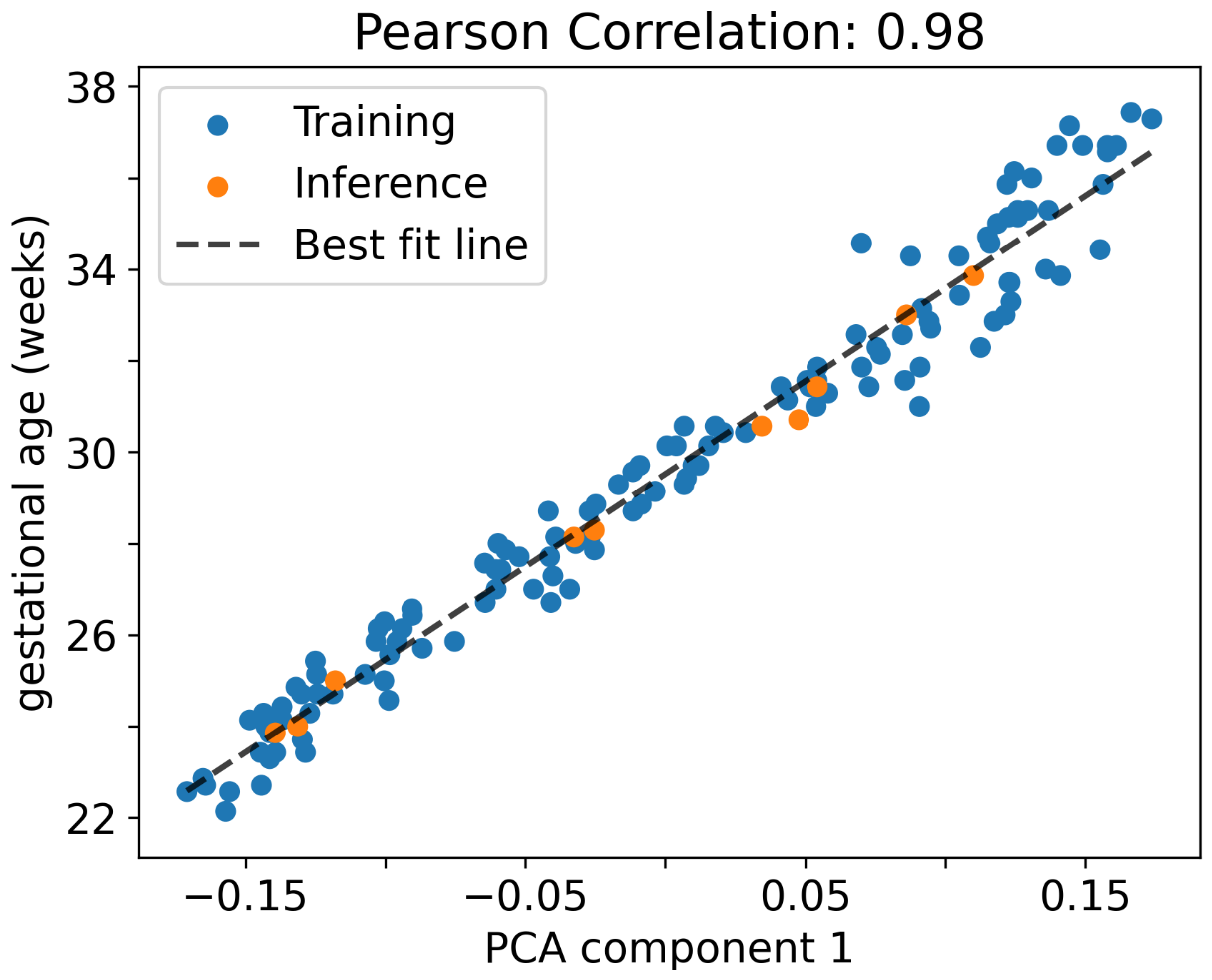
**Conditional Implicit Neural Atlas (CINA):**

- **represents** the fetal brain for the **entire time trajectory.**
- generates atlases **continuous in time and space.**
- models **neurotypical** and **pathological** brains by anatomical **conditioning.**

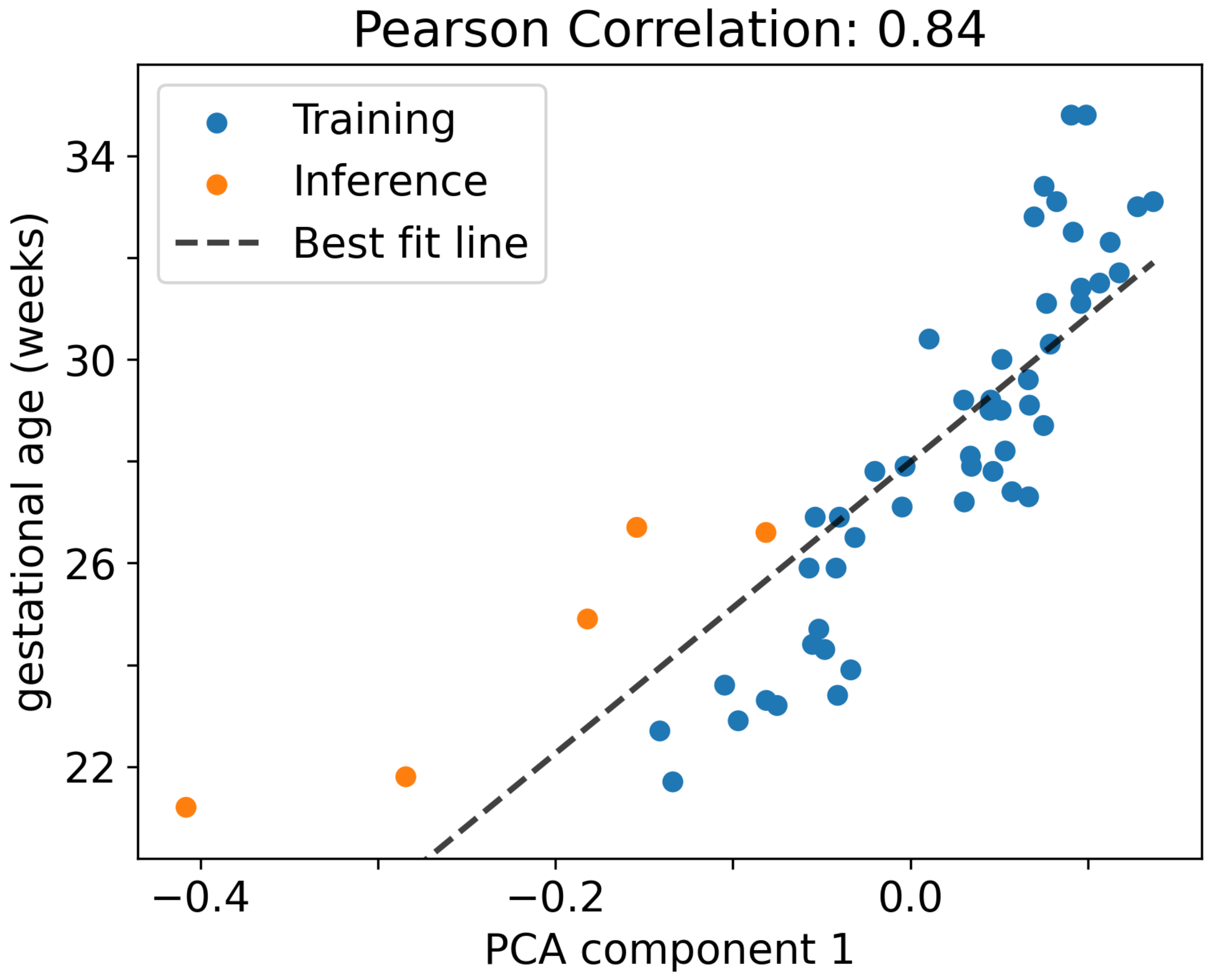
# QUESTIONS

# Latent Representation

dHCP Healthy




FeTA Healthy + Pathological




FeTA Healthy + Pathological w **e/c**